

Abtin Olaee

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EDUCATION

Masters of Science, Data Science | Expected May 2027

San Jose State University

Bachelor of Science, Statistics and Data Science | Jun 2025

University of California Santa Barbara

Dean's Honors (L&S) - Fall 2023, Spring 2024, Spring 2025

RELEVANT COURSEWORK

CS 271 - Topics in Machine Learning

MATH 164 - Mathematical Statistics

MATH 250 - Mathematical Data Visualization

MATH 252 - Cluster Analysis

MATH 261A - Regression Theory and Methods

MATH 261B - Design and Analysis of Experiments

WORK EXPERIENCE

Graduate Research Intern | Sep 2025 - Present

Wildfire Interdisciplinary Research Center of SJSU

- Conducted exploratory analysis of large-scale weather data
- Built Python ETL pipelines (xarray, h5netcdf) for analysis ready datasets
- Produced topography-driven visualizations;

Software Engineer Intern | Jun 2023 - Aug 2023

Frugal Innovation Hub of Santa Clara University

- Developed a responsive English-as-a-Second-Language (ESL) learning application using Firebase and Flutter
- Collaborated to ensure data integrity and UI/UX excellence
- Delivered milestone presentations and demos to supervising faculty, focusing on project goals and measurable outcomes

PROJECTS

AI Weather Framework for Wildfire Applications | Current

- Evaluated the temporal and spatial accuracy of six AI and NWP models using large-scale wind forecasts and Synoptic weather-station observations
- Built Python ETL pipelines with xarray and h5netcdf to clean, align, and standardize heterogeneous forecast and observation data into analysis-ready NetCDF datasets
- Developed a forecast-verification framework using RMSE, MAE, MAPE, bias, and spatial correlation to compare across initialization times, forecast lead times, and operational verification windows
- Created terrain-informed geospatial visualizations to identify how topography, station location, and localized weather conditions influenced forecast errors
- First-authored a research manuscript on AI weather-model performance and submitted it for peer review

Hyperspectral Vegetation Classification | May 2026

- Analyzed a 430-band hyperspectral vegetation dataset using spectral profiling, correlation analysis, and PCA to identify redundancy and informative wavelengths for land-cover classification
- Built a memory-efficient Python pipeline using Zarr, NumPy, rasterio, and scikit-learn to process large imagery, generate stratified training samples, and classify water, soil, canopy, and grass
- Reduced the feature set from 430 to 7 wavelength bands while maintaining 92.7% classification accuracy, accelerating Random Forest training by 13.6× and full-image prediction by 2.9×

Fire Weather and Nocturnal Drying Analysis of the 2020 SCU and CZU Lightning Complex Fires | Nov 2025

- Analyzed surface-station and upper-air observations from the 2020 SCU and CZU Lightning Complex fires to identify nocturnal drying and critical fire-weather conditions
- Calculated and compared the Fosberg, Hot-Dry-Windy, and Santa Ana wildfire indices using R to evaluate periods of elevated fire-spread potential
- Created time-series, vertical-profile, and station-level visualizations showing how elevation, terrain, humidity, and wind influenced wildfire risk across inland and coastal environments

Project DIANA | May 2023

- Received \$3,800 in IEEE funding
- Developed UAV software to enable spatial mapping of AQI and gas concentration
- Enabled real-time interpretation of atmospheric conditions utilizing Arduino system

UNIVERSITY EXPERIENCE

Vice President and Co-Founder | Nov 2021 - May 2023

IEEE Student Chapter, Ohlone College

Co-Founded the first IEEE branch at a California community college. Led a team of 50+ students and organized technical project showcases and conference participation. Software contributor to Project DIANA, creating UAV-based gas detection and real-time mapping systems

TEACHING AND MENTORING EXPERIENCE

Robotics Instructor | Jul 2025 - Jan 2026

Magikid Fremont

Computer Science Tutor | Sep 2022 - May 2023

Ohlone College

Assistant Wrestling Coach | Dec 2021 - Feb 2023

Irvington High School

SKILLS & CERTIFICATIONS

Languages & Tools: Python, R, Scikit-learn, Tensorflow, Xarray, NumPy/Pandas, SQL, Bash, Git, Linux/Unix, Matplotlib, Cartopy

Cloud Platforms: Google Cloud Platform, Apache Spark, AWS S3, Firebase, Databricks

Certifications: Microsoft Technology Associate - Security Fundamentals, NWCG S-190: Introduction to Wildland Fire Behavior